THE BLOWING UP INDUCES A BIJECTION ON THE GENERIC POINTS OUTSIDE THE CENTER

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 8000-10000 granted to this lane by CONTROL 8.83.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- The exceptional divisor is Cartier and the complement is an isomorphism. Lemma lemma-blowing-up-gives-effective-Cartier-divisor (8382-8395), used at proof lines 8549-8552. Reduces the claim to the Cartier divisor statement.
- Characterisation of effective Cartier divisors locally as V(f) with f a nonzero-divisor. Lemma lemma-characterize-effective-Cartier-divisor, used at proof lines 8553-8556. Produces the local equation f.
- A nonzerodivisor is not contained in a minimal prime of a reduced ring. Algebra, Lemma algebra-lemma-minimal-prime-reduced-ring, used at proof lines 8561-8563. Supplies the contradiction.

Source reference. Original lemma 00EU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 548–548. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02OS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3527–3527. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01WS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2493–2493. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let X be a scheme. Let $Z \subset X$ be a closed subscheme. Let $b : X' \to X$ be the blowing up of X along Z. Then b induces an bijective map from the set of generic points of irreducible components of X' to the set of generic points of irreducible components of X which are not in Z.*

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. The exceptional divisor $E \subset X'$ is an effective Cartier divisor and $X' \setminus E \to X \setminus Z$ is an isomorphism, see Lemma 02OS. Thus it suffices to show the following: given an effective Cartier divisor $D \subset S$ of a scheme S none of the generic points of irreducible components of S are contained in D. To see this, we may replace S by the members of an affine open covering. Hence by Lemma 01WS we may assume $S = \mathrm{Spec}(A)$ and $D = V(f)$ where $f \in A$ is a nonzerodivisor. Then we have to show f is not contained in any minimal prime ideal $\mathfrak{p} \subset A$. If so, then f would map to a nonzerodivisor contained in the maximal ideal of $R_\mathfrak{p}$ which is a contradiction with Algebra, Lemma 00EU. $\square$